



Option : Science Informatique Option Sciences Informatique

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ÉTUDIANT EN : 4^{ème} années

DEVOIR : Analyse et modélisation

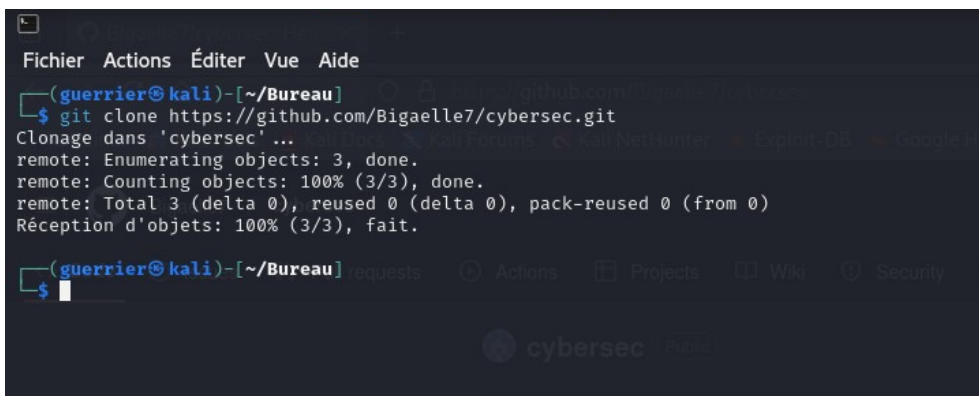
PRESENTE AU PROF : Ismael SAINT AMOUR

SUJET : Travaux Dirigés Virtualisation de Kali Linux

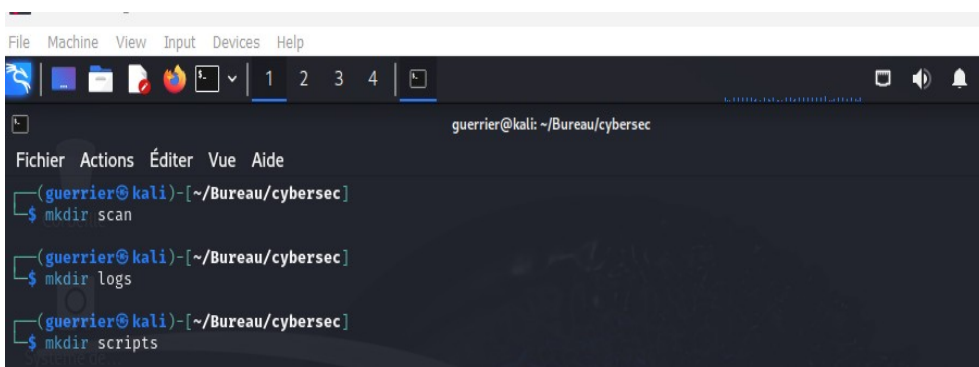
Pacot, le 10/02/25

Ce Travail est conçu la virtualisation de Kali Linux et l'apprentissage des premières commandes est de permet de familiariser avec l'environnement Kali Linux, de comprendre les bases de la virtualisation, et de maîtriser les commandes essentielles pour naviguer et interagir avec un système Linux.

1) Création d'un dossier cybersec avec trois sous-dossiers : scan , logs , scripts .

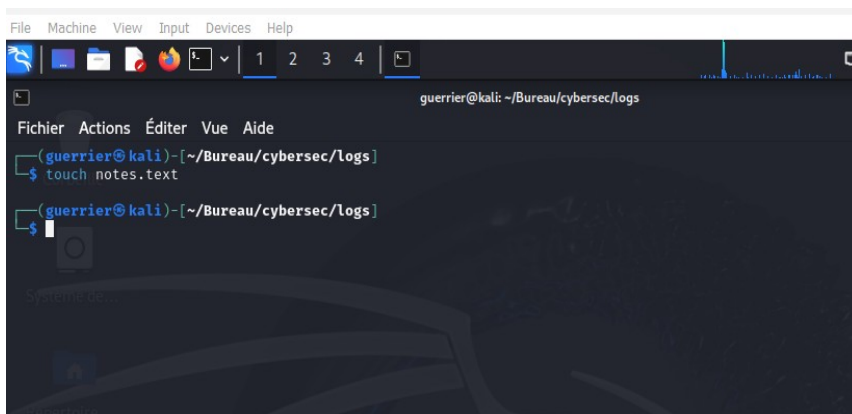


```
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau]
$ git clone https://github.com/Bigaelle7/cybersec.git
Clonage dans 'cybersec' ...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Réception d'objets: 100% (3/3), fait.
```



```
File Machine View Input Devices Help
guerrier@kali: ~/Bureau/cybersec
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec]
$ mkdir scan
(guerrier@kali)-[~/Bureau/cybersec]
$ mkdir logs
(guerrier@kali)-[~/Bureau/cybersec]
$ mkdir scripts
```

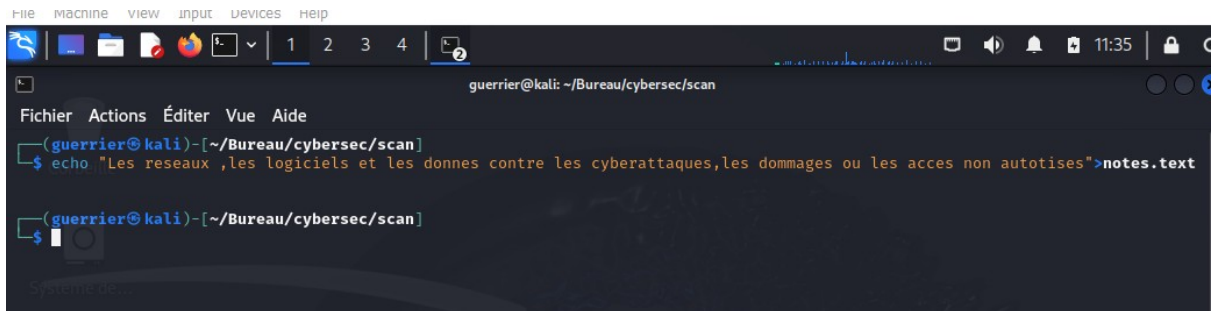
2) Ajout d'un fichier notes.txt dans scan et logs



```
File Machine View Input Devices Help
guerrier@kali: ~/Bureau/cybersec/logs
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec/logs]
$ touch notes.txt
(guerrier@kali)-[~/Bureau/cybersec/logs]
$
```

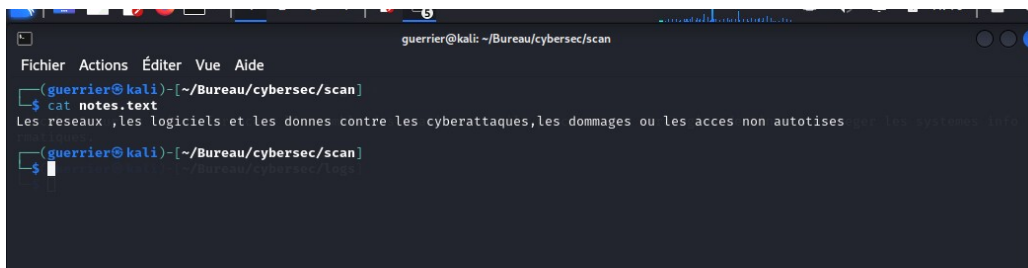
3) Ajout du contenu dans les fichiers textes (notes.txt), puis affichez le contenu des fichiers.

Ajout du contenu dans le sous dossier scan



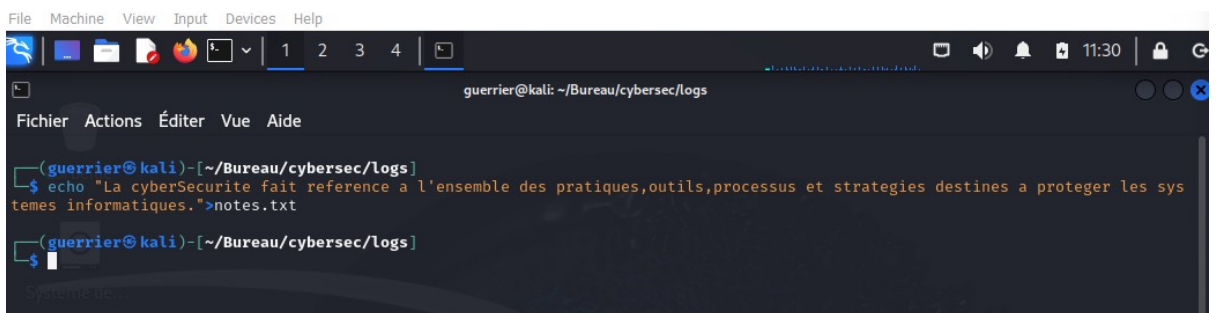
```
guerrier@kali: ~/Bureau/cybersec/scan
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec/scan]
$ echo "Les reseaux ,les logiciels et les donnees contre les cyberattaques,les dommages ou les acces non autorises">notes.txt
(guerrier@kali)-[~/Bureau/cybersec/scan]
$
```

Affichage du contenu



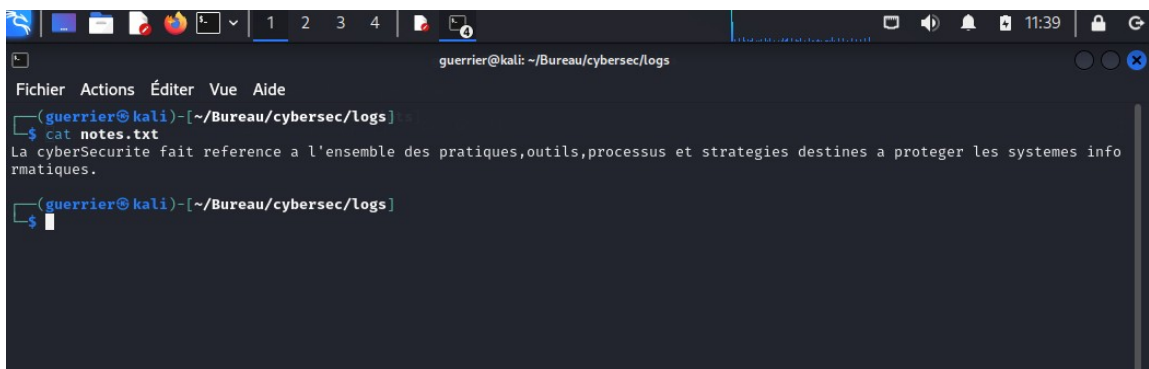
```
guerrier@kali: ~/Bureau/cybersec/scan
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec/scan]
$ cat notes.txt
Les reseaux ,les logiciels et les donnees contre les cyberattaques,les dommages ou les acces non autorises
(guerrier@kali)-[~/Bureau/cybersec/scan]
$
```

Ajout du contenu dans le sous dossier logs



```
guerrier@kali: ~/Bureau/cybersec/logs
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec/logs]
$ echo "La cyberSecurite fait reference a l'ensemble des pratiques,outils,processus et strategies destines a proteger les sys
temes informatiques.">notes.txt
(guerrier@kali)-[~/Bureau/cybersec/logs]
$
```

Affichage du contenu



```
guerrier@kali: ~/Bureau/cybersec/logs
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec/logs]
$ cat notes.txt
La cyberSecurite fait reference a l'ensemble des pratiques,outils,processus et strategies destines a proteger les systemes info
rmatiques.
(guerrier@kali)-[~/Bureau/cybersec/logs]
$
```

- 4) Copie du fichier (notes.txt) dans le sous-dossier scripts
Et vérification si le fichiers a été copié.

```
guerrier@kali: ~/Bureau/cybersec
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec]
$ cp ~/Bureau/cybersec/scan/notes.text ~/Bureau/cybersec/scripts/

(guerrier@kali)-[~/Bureau/cybersec]
$ ls -l
total 16
drwxrwxr-x 2 guerrier guerrier 4096  7 fév 11:32 logs
-rw-rw-r-- 1 guerrier guerrier   22  7 fév 10:45 README.md
drwxrwxr-x 2 guerrier guerrier 4096  7 fév 12:15 scan
drwxrwxr-x 2 guerrier guerrier 4096  7 fév 12:24 scripts

(guerrier@kali)-[~/Bureau/cybersec]
$ ls -l ~/Bureau/cybersec/scripts/
total 4
-rw-rw-r-- 1 guerrier guerrier 106  7 fév 12:24 notes.text

(guerrier@kali)-[~/Bureau/cybersec]
$
```

5) Déplacement du fichier (notes.txt) dans le sous-dossier scan .

```
guerrier@kali: ~/Bureau/cybersec/scripts
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec/scripts]
$ mv ~/Bureau/cybersec/scripts/notes.text ~/Bureau/cybersec/scan/

(guerrier@kali)-[~/Bureau/cybersec/scripts]
$ ls -l
total 0
/Bureau/cybersec/scan/notes.text ~/Bureau/cybersec/scripts/

(guerrier@kali)-[~/Bureau/cybersec/scripts]
$
```

6) Suppression du fichier (notes.txt) dans le sous-dossier scripts et vérification si le fichiers a été supprimé.

```
File Machine View Input Devices Help
guerrier@kali: ~/Bureau/cybersec/scripts
Fichier Actions Éditer Vue Aide
(guerrier@kali)-[~/Bureau/cybersec/scripts]
$ rm notes.text

(guerrier@kali)-[~/Bureau/cybersec/scripts]
$ ls -l
total 0

(guerrier@kali)-[~/Bureau/cybersec/scripts]
$
```

7) Supprimez les sous-dossiers : scan , logs , scripts et vérification si les vérifier si les sous-dossiers ont été supprimés.

```
File Machine View Input Devices Help
guerrier@kali: ~/Bureau/cybersec

Fichier Actions Éditer Vue Aide

(guerrier@kali)-[~/Bureau/cybersec/scripts]
$ rm -r scan

(guerrier@kali)-[~/Bureau/cybersec/scripts]
$ rm -r logs

(guerrier@kali)-[~/Bureau/cybersec/scripts]
$ rm -r scripts

(guerrier@kali)-[~/Bureau/cybersec]
$ ls -l
total 4
-rw-rw-r-- 1 guerrier guerrier 22  7 fév 10:45 README.md

(guerrier@kali)-[~/Bureau/cybersec]
$ ls
README.md

(guerrier@kali)-[~/Bureau/cybersec]
$
```

8) Scannerisation d'un réseau :

ip a

```
guerrier@kali: ~/Bureau/cybersec

Fichier Actions Éditer Vue Aide

(guerrier@kali)-[~/Bureau/cybersec/scripts]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b0:99:f6 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 80390sec preferred_lft 80390sec
    inet6 fe80::a00:27ff:feb0:99f6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Utilisation nmap

```

(guerrier@kali)~[~/Bureau/cybersec]
$ nmap -sv 192.168.1.1 /Bureau/cybersec/scripts
Nmap 7.94SVN ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}

TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file

HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PY/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host

SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan

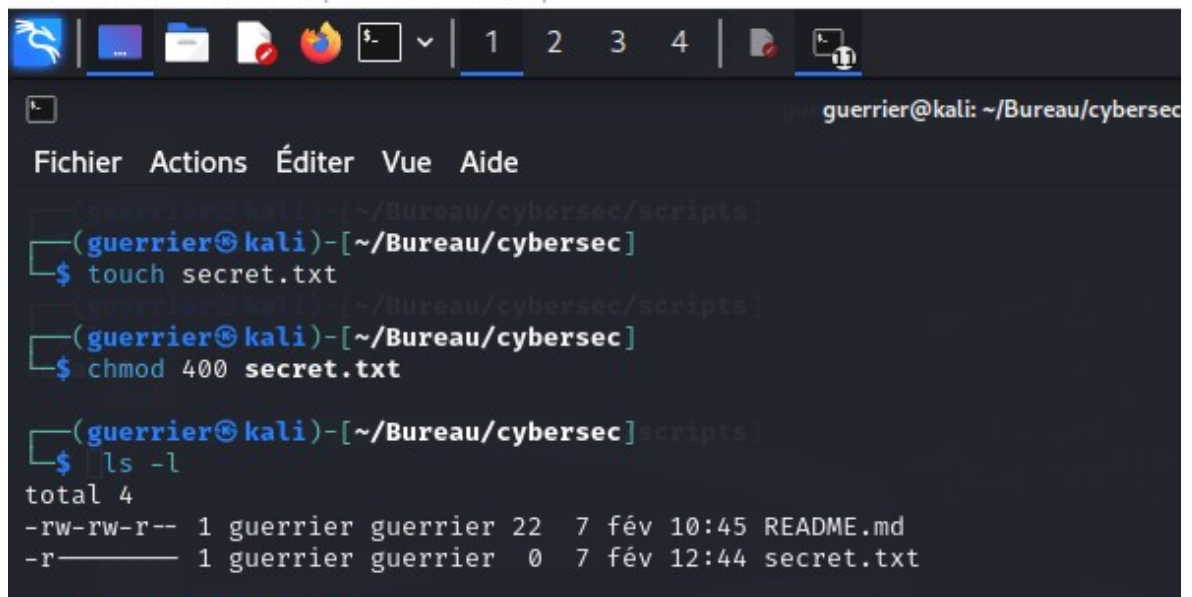
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>

SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)

```

9) Manipulation les permissions :

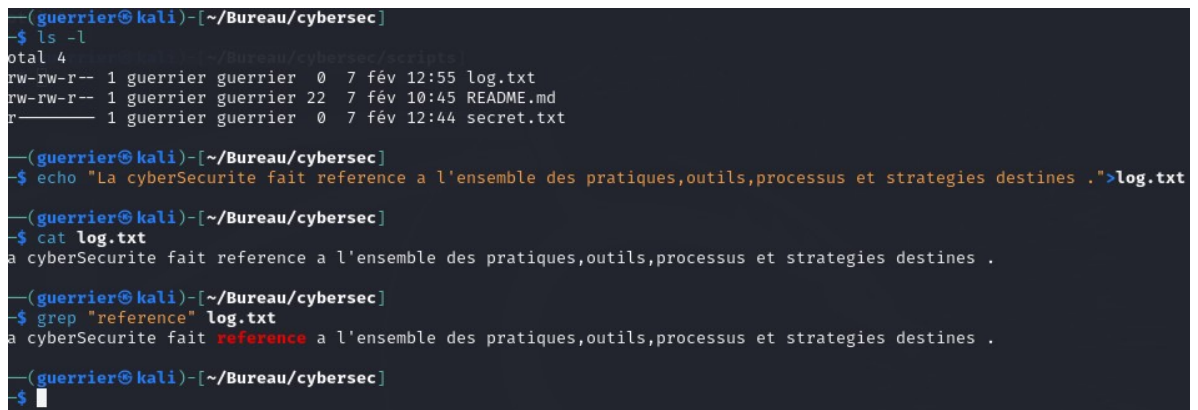
Création d'un fichier secret.txt et change ses permissions pour qu'il ne soit accessible qu'en lecture par le propriétaire



The screenshot shows a terminal window with a dark background. At the top, there is a menu bar with 'Fichier', 'Actions', 'Éditer', 'Vue', and 'Aide'. Below the menu bar, the terminal prompt is '(guerrier@kali) - [~/Bureau/cybersec]'. The user enters the command 'touch secret.txt', followed by 'chmod 400 secret.txt'. Then, the user enters 'ls -l' in the same directory. The output of 'ls -l' shows a directory listing with permissions, owner, group, size, date, and filename. The files listed are 'README.md' and 'secret.txt'.

```
(guerrier@kali) - [~/Bureau/cybersec]
$ touch secret.txt
(guerrier@kali) - [~/Bureau/cybersec]
$ chmod 400 secret.txt
(guerrier@kali) - [~/Bureau/cybersec]
$ ls -l
total 4
-rw-rw-r-- 1 guerrier guerrier 22 7 fév 10:45 README.md
-r----- 1 guerrier guerrier 0 7 fév 12:44 secret.txt
```

10) Utilisation grep :



The screenshot shows a terminal window with a dark background. The user enters 'ls -l' and the output shows a directory listing with 'log.txt', 'README.md', and 'secret.txt'. Then, the user enters 'echo "La cyberSecurite fait reference a l'ensemble des pratiques,outils,processus et strategies destines .">log.txt'. Next, the user enters 'cat log.txt' and the output shows the content of 'log.txt'. Finally, the user enters 'grep "reference" log.txt' and the output shows the line 'La cyberSecurite fait reference a l'ensemble des pratiques,outils,processus et strategies destines .'.

```
(guerrier@kali) - [~/Bureau/cybersec]
$ ls -l
total 4
-rw-rw-r-- 1 guerrier guerrier 0 7 fév 12:55 log.txt
-rw-rw-r-- 1 guerrier guerrier 22 7 fév 10:45 README.md
-r----- 1 guerrier guerrier 0 7 fév 12:44 secret.txt
(guerrier@kali) - [~/Bureau/cybersec]
$ echo "La cyberSecurite fait reference a l'ensemble des pratiques,outils,processus et strategies destines .">log.txt
(guerrier@kali) - [~/Bureau/cybersec]
$ cat log.txt
La cyberSecurite fait reference a l'ensemble des pratiques,outils,processus et strategies destines .
(guerrier@kali) - [~/Bureau/cybersec]
$ grep "reference" log.txt
La cyberSecurite fait reference a l'ensemble des pratiques,outils,processus et strategies destines .
(guerrier@kali) - [~/Bureau/cybersec]
$
```

11) Les commandes :

```
File Machine View Input Devices Help
guerrier@kali: ~/Bureau/cybersec

Fichier Actions Éditer Vue Aide

guerrier@kali: ~/Bureau/cybersec/scripts
(guerrier@kali)-[~/Bureau/cybersec]
$ df -h
Sys. de fichiers Taille Utilisé Dispo Uti% Monté sur
udev                926M      0  926M   0% /dev
tmpfs               198M    996K  197M   1% /run
/dev/sda1           38G      16G   20G  46% /
tmpfs               988M    4,0K  988M   1% /dev/shm
tmpfs               5,0M      0   5,0M   0% /run/lock
tmpfs              1,0M      0   1,0M   0% /run/credentials/systemd-journald.service
tmpfs              988M    8,0K  988M   1% /tmp
tmpfs              1,0M      0   1,0M   0% /run/credentials/getty@tty1.service
tmpfs              198M    120K  198M   1% /run/user/1000

(guerrier@kali)-[~/Bureau/cybersec]
$ du -sh
200K .

(guerrier@kali)-[~/Bureau/cybersec]
$ free -h
              total        utilisé        libre        partagé    tamp/cache    disponible
Mem:          1,9Gi         1,4Gi         133Mi          85Mi         605Mi         510Mi
Échange:       2,0Gi         131Mi         1,9Gi
```

```
(guerrier@kali)-[~/Bureau/cybersec]
$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.6 23044 14124 ?        Ss   10:54   0:01 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    10:54   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    10:54   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   10:54   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   10:54   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   10:54   0:00 [kworker/R-slub_flushwq]
root           7  0.0  0.0      0     0 ?        I<   10:54   0:00 [kworker/R-netns]
root           8  0.0  0.0      0     0 ?        I    10:54   0:06 [kworker/0:0-events]
root          12  0.0  0.0      0     0 ?        I<   10:54   0:00 [kworker/R-mm_percpu_wq]
root          13  0.0  0.0      0     0 ?        I    10:54   0:00 [rcu_tasks_kthread]
root          14  0.0  0.0      0     0 ?        I    10:54   0:00 [rcu_tasks_rude_kthread]
root          15  0.0  0.0      0     0 ?        I    10:54   0:00 [rcu_tasks_trace_kthread]
root          16  0.0  0.0      0     0 ?        S    10:54   0:02 [ksoftirqd/0]
root          17  0.0  0.0      0     0 ?        I    10:54   0:01 [rcu_preempt]
root          18  0.0  0.0      0     0 ?        S    10:54   0:00 [rcu_exp_par_gp_kthread_worker/
root          19  0.0  0.0      0     0 ?        S    10:54   0:00 [rcu_exp_gp_kthread_worker]
root          20  0.0  0.0      0     0 ?        S    10:54   0:00 [migration/0]
root          21  0.0  0.0      0     0 ?        S    10:54   0:00 [idle_inject/0]
```

```
(guerrier@kali)-[~/Bureau/cybersec]
$ lspci
0:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
0:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
0:01.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
0:02.0 VGA compatible controller: VMware SVGA II Adapter
0:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)
0:04.0 System peripheral: InnoTek Systemberatung GmbH VirtualBox Guest Service
0:05.0 Multimedia audio controller: Intel Corporation 82801AA AC'97 Audio Controller (rev 01)
0:06.0 USB controller: Apple Inc. KeyLargo/Intrepid USB
0:07.0 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
0:0b.0 USB controller: Intel Corporation 82801FB/FBM/FR/FW/FRW (ICH6 Family) USB2 EHCI Controller
0:0d.0 SATA controller: Intel Corporation 82801HM/HEM (ICH8M/ICH8M-E) SATA Controller [AHCI mode] (rev 02)
```

```
(guerrier@kali) [~/Bureau/cybersec]
$ sudo apt install traceroute
[sudo] Mot de passe de guerrier :
traceroute est déjà la version la plus récente (1:2.1.6-1).
Vous pouvez lancer « apt --fix-broken install » pour corriger ces problèmes.
Unsatisfied dependencies:
```

```
(guerrier@kali) [~/Bureau/cybersec]
$ traceroute google.com
traceroute to google.com (142.250.169.101): 30 hops, 254 bytes, 30.00 ms
 0  10.0.2.2 (10.0.2.2)  35.123 ms
 1  * * *
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
```

```
(guerrier@kali) [~/Bureau/cybersec]
$ netstat -tuln
Connexions Internet actives (seulement serveurs)
Proto Recv-Q Send-Q Adresse locale Adresse distante Etat
udp 0 0 0.0.0.0:40923 0.0.0.0:*
udp 0 0 10.0.2.15:3702 0.0.0.0:*
udp 0 0 239.255.255.250:3702 0.0.0.0:*
udp6 0 0 fe80::a00:27ff:feb:3702 :::*
udp6 0 0 ff02::c:3702 :::*
udp6 0 0 :::37504 :::*

(guerrier@kali) [~/Bureau/cybersec]
$ ss -tuln
State Recv-Q Send-Q Local Address:Port Peer Address:Port
udp UNCONN 0 0 0.0.0.0:40923 0.0.0.0:*
udp UNCONN 0 0 10.0.2.15:3702 0.0.0.0:*
udp UNCONN 0 0 239.255.255.250:3702 0.0.0.0:*
udp UNCONN 0 0 [fe80::a00:27ff:feb:99f6]:::3702 [::]:*
udp UNCONN 0 0 [ff02::c]:::3702 [::]:*
udp UNCONN 0 0 *:37504 *:*
```

```

(guerrier@kali)-[~/Bureau/cybersec]
└─$ journalctl
fév 03 15:09:25 kali kernel: Linux version 6.11.2-amd64 (dev@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-6) 14.2.0, GN>
fév 03 15:09:25 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.11.2-amd64 root=UUID=c2827348-3b47-4145-9500-290138e8be4>
fév 03 15:09:25 kali kernel: [Firmware Bug]: TSC doesn't count with P0 frequency!
fév 03 15:09:25 kali kernel: BIOS-provided physical RAM map:
fév 03 15:09:25 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
fév 03 15:09:25 kali kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
fév 03 15:09:25 kali kernel: BIOS-e820: [mem 0x00000000000f0000-0x000000000000ffffff] reserved
fév 03 15:09:25 kali kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000007ffeffff] usable
fév 03 15:09:25 kali kernel: BIOS-e820: [mem 0x0000000007fff0000-0x0000000007ffffff] ACPI data
fév 03 15:09:25 kali kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
fév 03 15:09:25 kali kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
fév 03 15:09:25 kali kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
fév 03 15:09:25 kali kernel: NX (Execute Disable) protection: active
fév 03 15:09:25 kali kernel: APIC: Static calls initialized
fév 03 15:09:25 kali kernel: SMBIOS 2.5 present.
fév 03 15:09:25 kali kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
fév 03 15:09:25 kali kernel: DMI: Memory slots populated: 0/0
fév 03 15:09:25 kali kernel: Hypervisor detected: KVM
fév 03 15:09:25 kali kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
fév 03 15:09:25 kali kernel: kvm-clock: using sched offset of 1818495400455 cycles
fév 03 15:09:25 kali kernel: clocksource: kvm-clock: mask: 0xffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 88159059>
fév 03 15:09:25 kali kernel: tsc: Detected 2095.990 MHz processor
fév 03 15:09:25 kali kernel: e820: update [mem 0x00000000-0x00000fff] usable => reserved
fév 03 15:09:25 kali kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
fév 03 15:09:25 kali kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
fév 03 15:09:25 kali kernel: MTRRs disabled by BIOS
fév 03 15:09:25 kali kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
fév 03 15:09:25 kali kernel: found SMP MP-table at [mem 0x0009fff0-0x0009ffff]

```

```

(guerrier@kali)-[~/Bureau/cybersec]
└─$ journalctl -f
fév 07 13:09:40 kali systemd[1]: Finished phpsessionclean.service - Clean php session files.
fév 07 13:13:36 kali sudo[71252]: guerrier : TTY=pts/16 ; PWD=/home/guerrier/Bureau/cybersec ; USER=root ; COMMAND=/usr/bin/apt
install traceroute
fév 07 13:13:36 kali sudo[71252]: pam_unix(sudo:session): session opened for user root(uid=0) by guerrier(uid=1000)
fév 07 13:13:39 kali sudo[71252]: pam_unix(sudo:session): session closed for user root
fév 07 13:15:01 kali CRON[72176]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
fév 07 13:15:01 kali CRON[72177]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
fév 07 13:15:01 kali CRON[72176]: pam_unix(cron:session): session closed for user root
fév 07 13:17:01 kali CRON[73162]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
fév 07 13:17:01 kali CRON[73163]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
fév 07 13:17:01 kali CRON[73162]: pam_unix(cron:session): session closed for user root

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(guerrier@kali)-[~/Bureau/cybersec]
└─$ journalctl -b
fév 07 10:54:50 kali kernel: Linux version 6.11.2-amd64 (dev@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-6) 14.2.0, GN>
fév 07 10:54:50 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.11.2-amd64 root=UUID=c2827348-3b47-4145-9500-290138e8be4>
fév 07 10:54:50 kali kernel: [Firmware Bug]: TSC doesn't count with P0 frequency!
fév 07 10:54:50 kali kernel: BIOS-provided physical RAM map:
fév 07 10:54:50 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
fév 07 10:54:50 kali kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
fév 07 10:54:50 kali kernel: BIOS-e820: [mem 0x00000000000f0000-0x000000000000ffffff] reserved
fév 07 10:54:50 kali kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000007ffeffff] usable
fév 07 10:54:50 kali kernel: BIOS-e820: [mem 0x0000000007fff0000-0x0000000007ffffff] ACPI data
fév 07 10:54:50 kali kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
fév 07 10:54:50 kali kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
fév 07 10:54:50 kali kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
fév 07 10:54:50 kali kernel: NX (Execute Disable) protection: active
fév 07 10:54:50 kali kernel: APIC: Static calls initialized
fév 07 10:54:50 kali kernel: SMBIOS 2.5 present.
fév 07 10:54:50 kali kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
fév 07 10:54:50 kali kernel: DMI: Memory slots populated: 0/0
fév 07 10:54:50 kali kernel: Hypervisor detected: KVM
fév 07 10:54:50 kali kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
fév 07 10:54:50 kali kernel: kvm-clock: using sched offset of 3767747351453 cycles
fév 07 10:54:50 kali kernel: clocksource: kvm-clock: mask: 0xffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 88159059>
fév 07 10:54:50 kali kernel: tsc: Detected 2095.990 MHz processor
fév 07 10:54:50 kali kernel: e820: update [mem 0x00000000-0x00000fff] usable => reserved
fév 07 10:54:50 kali kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
fév 07 10:54:50 kali kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
fév 07 10:54:50 kali kernel: MTRRs disabled by BIOS

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(guerrier@kali)~[/Bureau/cybersec]$ scripts
$ journalctl -n 10
fév 07 13:13:39 kali sudo[71252]: pam_unix(sudo:session): session closed for user root
fév 07 13:15:01 kali CRON[72176]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
fév 07 13:15:01 kali CRON[72177]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
fév 07 13:15:01 kali CRON[72176]: pam_unix(cron:session): session closed for user root
fév 07 13:17:01 kali CRON[73162]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
fév 07 13:17:01 kali CRON[73163]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
fév 07 13:17:01 kali CRON[73162]: pam_unix(cron:session): session closed for user root
fév 07 13:25:01 kali CRON[77159]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
fév 07 13:25:01 kali CRON[77160]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
fév 07 13:25:01 kali CRON[77159]: pam_unix(cron:session): session closed for user root

(guerrier@kali)~[/Bureau/cybersec]$ date
ven 07 fév 2025 13:28:08 EST

(guerrier@kali)~[/Bureau/cybersec]$ timedatectl
          Local time: ven 2025-02-07 13:28:31 EST
          Universal time: ven 2025-02-07 18:28:31 UTC
             RTC time: ven 2025-02-07 18:28:31
            Time zone: America/Port-au-Prince (EST, -0500)
System clock synchronized: no
              NTP service: inactive
          RTC in local TZ: no

(guerrier@kali)~[/Bureau/cybersec]$ hostnamectl
Static hostname: kali
          Icon name: computer-vm
         Chassis: vm
       Machine ID: 062ee443105147448471246bf1c15f8a
          Boot ID: c250a1668ad54412a146e95d897ffcdb
    Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
           Kernel: Linux 6.11.2-amd64
    Architecture: x86-64
   Hardware Vendor: innotek GmbH
   Hardware Model: VirtualBox
Firmware Version: VirtualBox
   Firmware Date: Fri 2006-12-01
   Firmware Age: 18y 2month 1w 1d

```

En conclusion, ce travail me permet de maîtriser les commandes essentielles pour naviguer et interagir avec un système Linux et me permet aussi de bien maîtriser le git hub.